Removing the Extra Smoothing Factor in Fackler's Kato–Beurling Converse

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Source and Target

The source paper is Stephan Fackler, *Regularity of Semigroups via the Asymptotic Behaviour at Zero*, arXiv:1203.5498. Corollary 2.5 proves that if a C_0-semigroup $(T(t))_{t\geq 0}$ is holomorphic, then for all large N one has

$$\limsup_{t\downarrow 0} \left\| (T(t) - \mathrm{Id})^N T(t)^K \right\|^{1/N} < 2$$

after choosing a sufficiently large integer K. Remark 2.6 observes that, for a bounded holomorphic semigroup on a Hilbert space whose negative generator has a bounded H^∞-calculus, the extra factor $T(t)^K$ can be omitted, and asks whether this remains true in general.

Moreover, taking $f(z) = (z-1)^N$ for some $N \in \mathbb{N}$ we obtain a variant of Kato's original result which is invariant under equivalent renorming of the underlying Banach space.

Corollary 2.5. *Let $(T(t))_{t\geq 0}$ be a C_0-semigroup on a Banach space X.*

(a) *$(T(t))_{t\geq 0}$ extends to a holomorphic semigroup if for some $N \in \mathbb{N}$*

$$\limsup_{t\downarrow 0} \left\| (T(t) - \mathrm{Id})^N \right\|^{1/N} < 2.$$

(b) *Conversely, if $(T(t))_{t\geq 0}$ is holomorphic, there exists $N_0 \in \mathbb{N}$ such that for every $N \geq N_0$ there exists $K_0 \in \mathbb{R}_+$ such that*

$$\limsup_{t\downarrow 0} \left\| (T(t) - \mathrm{Id})^N T(t)^K \right\|^{1/N} < 2 \quad \textit{for all } \mathbb{N} \ni K \geq K_0.$$

Remark 2.6 (On the additional factor of z^K). Let $X = H$ be a Hilbert space and $(T(z))_{z\in\Sigma}$ a bounded holomorphic semigroup on H whose negative generator has a bounded $H^\infty(\Sigma_\varphi)$-calculus ($\varphi \in (0, \pi]$). For the definition of H^∞-calculi and their properties we refer to [KW04, Section 9]. Then by [KW04, Theorem 11.13] there exists an equivalent scalar product $(\cdot|\cdot)_2$ on H for which $(T(t))_{t\geq 0}$ is contractive. Now [Paz83, Corollary 5.8] shows $\limsup_{t\downarrow 0} \|T(t) - \mathrm{Id}\|_2 < 2$. Hence, for the original norm we obtain

$$\limsup_{t\downarrow 0} \left\| (T(t) - \mathrm{Id})^N \right\|^{1/N} < 2$$

if $N \in \mathbb{N}$ is chosen sufficiently large. So in this case we can omit the additional factor of z^K. We do not know whether this is true in general.

This packet gives an affirmative answer. The proof does not use Hilbert space structure or an H^∞-calculus.

Proof Intuition

For a bounded holomorphic semigroup $T(t) = e^{-tA}$, the negative generator A is sectorial of some angle strictly smaller than $\pi/2$. Choose a larger sector angle $\psi < \pi/2$. On the two boundary rays of this sector, the scalar function $z \mapsto e^{-z} - 1$ is uniformly bounded by a number $q_\psi < 2$. Thus the scalar finite-difference function

$$f_N(z) = (e^{-z} - 1)^N$$

has boundary size at most q_ψ^N, apart from its harmless limit $(-1)^N$ at infinity. A sectorial contour estimate then gives

$$\|f_N(tA)\| \leq 1 + C \log(N+1) q_\psi^N + o(1)$$

uniformly for $t > 0$. Since $q_\psi < 2$, the N-th root is eventually strictly less than 2. Finally, an arbitrary holomorphic C_0-semigroup can be exponentially rescaled to a bounded holomorphic semigroup on a smaller sector, and this rescaling does not change the small-time N-th finite difference limsup.

Sectorial Estimate

We use the standard notation

$$\Sigma_\alpha = \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \alpha\}.$$

Recall that if $-A$ generates a bounded holomorphic C_0-semigroup, then A is sectorial of some angle $\theta < \pi/2$.

Lemma 1 (A boundary estimate for finite-limit functions)**.** *Let A be sectorial of angle $\theta < \pi/2$, and fix $\psi \in (\theta, \pi/2)$. There is a constant $C_{\psi,A}$ with the following property. Let f be holomorphic in a neighbourhood of $\overline{\Sigma_\psi}$, continuous at 0 and at infinity, with $f(0) = 0$ and $f(\infty) = L$. Put*

$$m_f(r) = \max_{\sigma=\pm 1} \left| f(re^{i\sigma\psi}) \right|, \qquad m_f^\infty(r) = \max_{\sigma=\pm 1} \left| f(re^{i\sigma\psi}) - L \right|.$$

Then for every $R \geq 1$,

$$\|f(A)\| \leq |L| + C_{\psi,A} \left(\int_0^1 m_f(r) \, \frac{dr}{r} + \int_1^R m_f(r) \, \frac{dr}{r} + \int_R^\infty m_f^\infty(r) \, \frac{dr}{r} \right).$$

Proof. This is the elementary sectorial functional-calculus estimate obtained from the usual keyhole contour around Σ_θ. For completeness we spell out the point of the estimate. Let

$$M_{\psi,A} := \sup_{\lambda \notin \Sigma_\psi} \left\| \lambda(\lambda - A)^{-1} \right\| < \infty.$$

On the two rays $\lambda = re^{\pm i\psi}$, the resolvent contributes at most $M_{\psi,A} \, dr/r$. Since $f(0) = 0$, the small circular arc vanishes as its radius tends to 0. On the large arc, the constant part L contributes exactly $L\mathrm{Id}$, and the remainder is controlled by m_f^∞. Taking the inner radius to 0 and the outer radius to infinity gives the displayed bound. This is the standard extended sectorial calculus for functions with finite limits at 0 and ∞, with the norm estimate read directly from the same contour. □

Lemma 2 (Uniform finite-difference bound)**.** *Let $-A$ generate a bounded holomorphic C_0-semigroup on a Banach space. Then there exist $q < 2$ and $C > 0$ such that for every $N \geq 2$ and every $t > 0$,*

$$\left\| (e^{-tA} - \mathrm{Id})^N \right\| \leq 1 + C \left(\frac{1}{N} + \log(N+1) q^N + e^{-N} \right).$$

Consequently,

$$\limsup_{N \to \infty} \sup_{t > 0} \left\| (e^{-tA} - \mathrm{Id})^N \right\|^{1/N} < 2.$$

Proof. Choose $\psi \in (\theta, \pi/2)$, where θ is a sectoriality angle for A, and put

$$q := \sup_{r > 0} \left| 1 - e^{-re^{i\psi}} \right|.$$

Then $q < 2$: the displayed function is continuous in r, tends to 0 at $r = 0$, tends to 1 as $r \to \infty$, and cannot equal 2 for finite r because $\left| e^{-re^{i\psi}} \right| = e^{-r\cos\psi} < 1$.

Fix $N \geq 2$ and set $f_N(z) = (e^{-z} - 1)^N$. The operator $(e^{-tA} - \mathrm{Id})^N$ is $f_N(tA)$. The scaled operator tA has the same sectorial angle and the same resolvent bound $M_{\psi,A}$, so Lemma 1 applies with constants independent of t. Since $f_N(0) = 0$ and $f_N(\infty) = (-1)^N$, it remains only to estimate the three scalar integrals in Lemma 1.

For $0 < r \le 1$, the inequality

$$\left|1 - e^{-w}\right| \le |w| \qquad (\operatorname{Re} w \ge 0)$$

gives $m_{f_N}(r) \le r^N$, and hence

$$\int_0^1 m_{f_N}(r)\, \frac{dr}{r} \le \frac{1}{N}.$$

For $1 \le r \le R$, $m_{f_N}(r) \le q^N$, so

$$\int_1^R m_{f_N}(r)\, \frac{dr}{r} \le q^N \log R.$$

Finally let $c = \cos\psi > 0$ and choose $R = N/c$. If $r \ge R$ and $u = e^{-re^{i\psi}}$, then $|u| \le e^{-N}$, and

$$\left|(1-u)^N - 1\right| \le N\,|u|\,(1 + |u|)^{N-1} \le 2Ne^{-cr}.$$

The same estimate holds on the lower boundary ray. Thus

$$\int_R^\infty m^\infty_{f_N}(r)\, \frac{dr}{r} \le 2N \int_R^\infty e^{-cr}\, \frac{dr}{r} \le 2e^{-N}.$$

Substitution into Lemma 1 gives the asserted uniform bound after increasing C. Since $q < 2$, the N-th root of the right hand side is eventually strictly smaller than 2. □

Main Result

Theorem 1. *Let $(T(t))_{t\ge 0}$ be a holomorphic C_0-semigroup on a Banach space X. Then there exists $N \in \mathbb{N}$ such that*

$$\limsup_{t\downarrow 0} \left\|(T(t) - \mathrm{Id})^N\right\|^{1/N} < 2.$$

In particular, the factor $T(t)^K$ in Fackler's Corollary 2.5(b) can be omitted.

Proof. First suppose that $(T(t))$ is bounded holomorphic, and write $T(t) = e^{-tA}$. Lemma 2 gives a sequence of integers N for which

$$\sup_{t>0} \left\|(T(t) - \mathrm{Id})^N\right\|^{1/N} < 2,$$

and hence the required small-time limsup estimate.

For a general holomorphic C_0-semigroup, restrict the holomorphic extension to a slightly smaller sector. By the semigroup law and local boundedness near 0, there is $\omega \in \mathbb{R}$ such that

$$S(z) := e^{-\omega z} T(z)$$

is bounded holomorphic on that smaller sector. The bounded case gives an integer N with

$$\limsup_{t\downarrow 0} \left\|(S(t) - \mathrm{Id})^N\right\|^{1/N} < 2.$$

Since $T(t) = e^{\omega t} S(t)$, we have

$$T(t) - \mathrm{Id} = e^{\omega t}(S(t) - \mathrm{Id}) + (e^{\omega t} - 1)\mathrm{Id}.$$

For this fixed N, expanding the N-th power shows that

$$(T(t) - \mathrm{Id})^N - e^{N\omega t}(S(t) - \mathrm{Id})^N \to 0$$

in operator norm as $t \downarrow 0$, because $(S(t))$ is bounded near 0. As $e^{N\omega t} \to 1$, the desired strict inequality transfers from S to T. □

Verification and Novelty Notes

No numerical computation is used. The proof is an analytic sectorial-calculus argument. The main review point is Lemma 1; it is the standard contour estimate for sectorial operators, but it is where all operator-theoretic content is concentrated.

Local duplicate checks were run against `runs/fa_banach_001/registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` for arXiv id `1203.5498`, the phrases `additional factor`, `z^K`, `Kato-Beurling`, and the finite-difference expressions in Corollary 2.5. No completed existing packet for this exact question was found before this promotion.

A bounded web search on July 18, 2026 used the exact phrases `additional factor z^K Kato-Beurling`, `limsup (T(t)-Id)^N T(t)^K < 2`, `Pazy limsup T(t)-I holomorphic counterexample`, and the source paper title. It found adjacent literature, including Batty–Gomilko–Tomilov's Besov-calculus paper and Hytönen–Lappas's quantitative Kato estimates for bounded holomorphic semigroups, but no separate paper explicitly answering Fackler's Remark 2.6 by removing the z^K factor. Novelty confidence is `bounded-search likely new`; this is not a full bibliographic audit.

Limitations and Review Recommendation

The theorem answers the question as stated for holomorphic C_0-semigroups on arbitrary Banach spaces. It does not provide an explicit optimal N; the proof gives existence through the sectoriality angle and resolvent constant.

Human review should check:

- the contour estimate in Lemma 1, especially the treatment of the finite limit $f(\infty)$;
- the transfer from a general holomorphic semigroup to an exponentially rescaled bounded holomorphic semigroup;
- the scalar tail estimate for $(1 - e^{-re^{i\psi}})^N - 1$.

References

[1] S. Fackler, *Regularity of Semigroups via the Asymptotic Behaviour at Zero*, Semigroup Forum 87 (2013), 21–42; arXiv:1203.5498.

[2] M. Haase, *The Functional Calculus for Sectorial Operators*, Operator Theory: Advances and Applications 169, Birkhauser, 2006.

[3] C. J. K. Batty, A. Gomilko, and Y. Tomilov, *A Besov algebra calculus for generators of operator semigroups and related norm-estimates*, Mathematische Annalen 379 (2021), 23–93.

[4] T. Hytönen and S. Lappas, *Quantitative estimates for bounded holomorphic semigroups*, Semigroup Forum 108 (2024), 115–144.